CABINET: Content Relevance Based Noise Reduction for Table Question Answering

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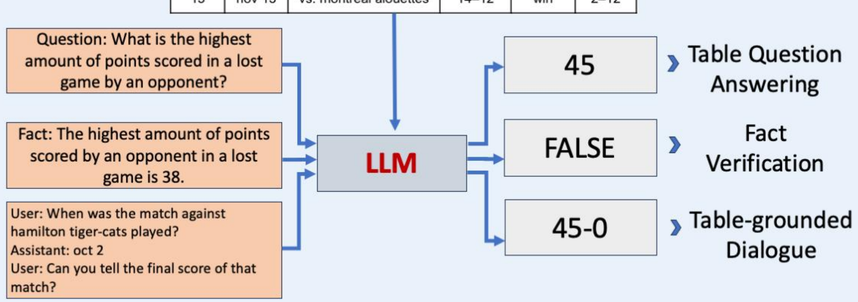
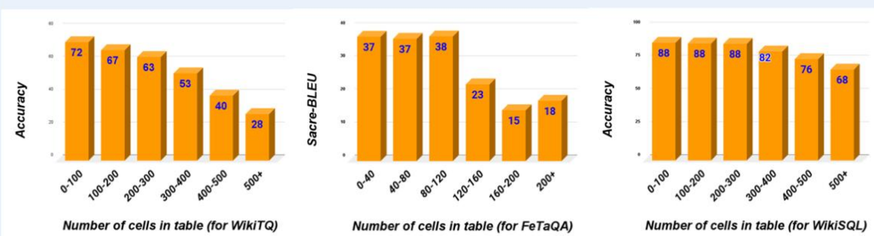
LLMs have been used for Table Understanding

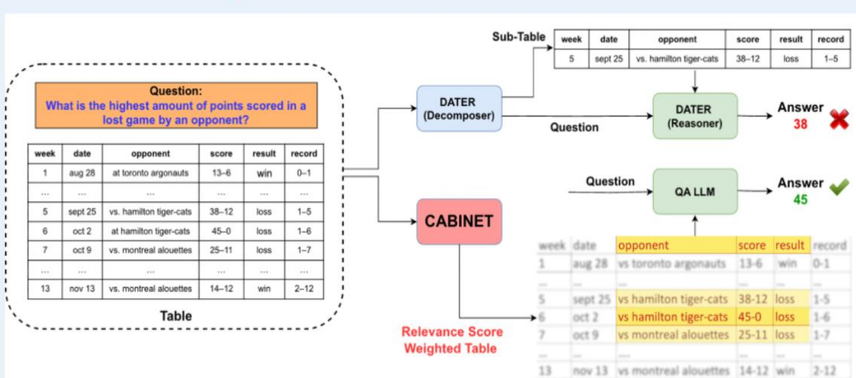
week	date	opponent	score	result	record
1	aug 28	at toronto argonauts	13–6	win	0–1
…	…	…	…	…	…
5	sept 25	vs. hamilton tiger-cats	38–12	loss	1–5
6	oct 2	at hamilton tiger-cats	45–0	loss	1–6
7	oct 9	vs. montreal alouettes	25–11	loss	1–7
…	…	…	…	…	…
13	nov 13	vs. montreal alouettes	14–12	win	2–12



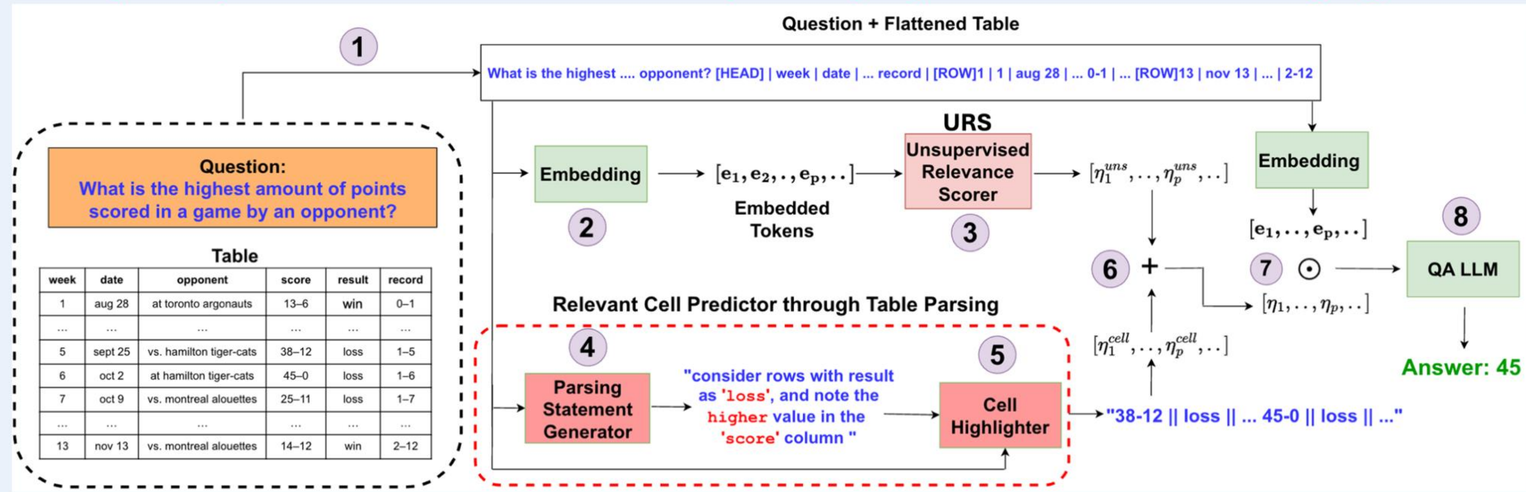
Large Tabular Data degrades LLM Performance



Hard Decomposition can cause Information Loss



Our Approach: CABINET assigns Relevance Score to Tabular Data Tokens through Two Modules - 1) Unsupervised Relevance Scorer and 2) Weakly Supervised Relevant Cell Highlighter



CABINET (despite having less #params) outperforms several SoTA Methods

Method	Acc.	#params
OmniTab	62.7	406M
FlanT5-xl	64.4	2.9B
DATER (GPT-3)	65.9	175B
CABINET	**69.1**	560M

WikiTQ

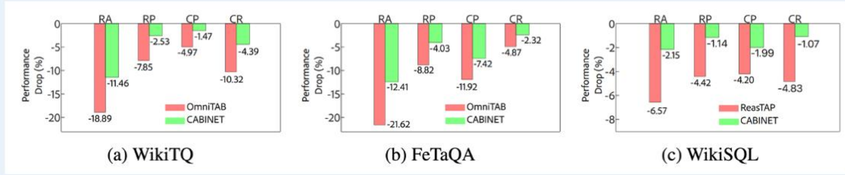
Method	S-BLEU	#params
TAPEX	34.7	406M
FlanT5-xl	36.2	2.9B
DATER (GPT-3)	30.9	175B
CABINET	**40.5**	560M

FeTaQA

Method	Acc.	#params
ReasTAP	88.8	406M
FlanT5-xl	87.8	2.9B
StructGPT	54.4	175B
CABINET	**89.5**	560M

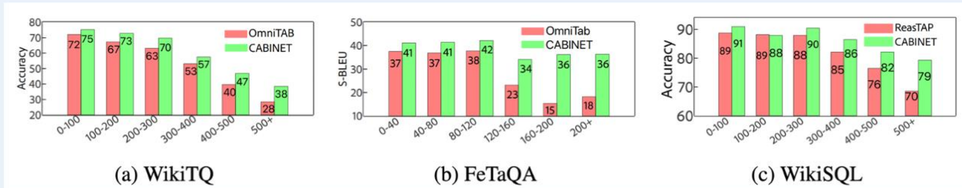
WikiSQL

CABINET is more Robust to Noise and Table Perturbations



RA – Row Addition
RP – Row Permutation
CP – Column Permutation
CR – Cell Replacement

CABINET is more Performant while Handling Irrelevant Data in Large Tables



Clustering URS Output Representations is Helpful

$\mathcal{L}_{clu}$	$\mathcal{L}_{sep}$	$\mathcal{L}_{sparse}$	WikiTQ	FeTaQA	WikiSQL
✗	✗	✗	60.8	35.1	86.2
✗	✗	✓	60.9	35.1	86.3
✓	✗	✗	62.7	35.0	88.9
✓	✗	✓	61.0	35.0	**89.5**
✓	✓	✗	61.0	35.1	89.1
✓	✓	✓	**65.6**	**35.8**	89.3

Using Both Modules gives Best Performance

λ_{uns}	λ_{cell}	WikiTQ	FeTaQA,	WikiSQL
1	0	65.6	35.8	**89.2**
0.7	0.3	**69.1**	**40.5**	**89.2**
0.5	0.5	68.6	**40.5**	88.9
0.3	0.7	67.0	38.9	88.8
0	1	37.6	24.2	34.1